Yasmine Zerai

AI Engineer | LLM Agents in Production, RAG & Agent Observability

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Profile

Final-year Software Engineering student at INSAT (Engineering Degree, Master's equivalent) with **1+ year of salaried production experience** at an IoT startup, held alongside a full course load. I build **LLM agents that run in production**, the **data and scoring models** they rely on, and the **full-stack platform** that puts them in front of clients. I also build and publish **loopview**, an open-source debugger for AI agents (**on PyPI**). Fluent in French and English. Looking for a **6-month end-of-studies internship (PFE)** in AI engineering from **January or February 2027**.

Professional Experience

AI & Data Science Engineer, Beekeeper Tech (IoT startup for bee colony monitoring) *08/2025 - Present*

- Built an **autonomous ReAct agent** (OpenAI function calling, 8 tools) that investigates colony anomalies every night, checking weather and **neighbouring hives** to separate environmental causes from colony problems. It **replaced the manual review completely** and is the team's **only source of truth for colony health**.
- Made the agent safe to trust: **schema-validated output** with self-correction, **fallback to standard alerts** when it fails, and **full tracing** of every tool call so any alert leads back to its evidence.
- Designed the **two-tier nightly alerting pipeline**: a rule engine checks every device at **zero model cost**, only biological anomalies escalate to the agent, and **four volume controls** (deduplication, severity priority, per-run cap, 3-day suppression) keep false alarms down.
- Improved the production **health scoring model** (0 to 10, with sub-scores): built a **season-aware weight assessment**, and a **brood presence confidence** score after finding in the data that many hives scored near zero **without being unhealthy**. Both now run **in production**.
- Prototyped the platform in Streamlit, then **rebuilt it and owned it end to end, from architecture to deployment**: **React** (Vite, Recharts), real-time **WebSocket** notifications, async **FastAPI**, **PostgreSQL** and MongoDB, fully bilingual English/Arabic.
- Designed a **RAG system** (LangChain, vector database, semantic chunking) for bilingual insights and PDF reports, and **ETL pipelines** for IoT time series (Pandas, Scikit-learn).
- Delivered it to **government and international organisation clients** on **2,000+ IoT devices** and presented it in **two product demos** (**Docker**, **AWS**, **MQTT**, **GitHub Actions**). Now building a **second client platform** (drone imagery, **PostGIS**).

Full-Stack Intern, then Web Development Instructor, Opus Lab *05/2025 - 08/2025*

- Delivered production modules on **Next.js / Nest.js / MongoDB** (TypeScript) and set up **QA** where none existed, then was **invited back to teach a 4-week bootcamp** to **15+ junior developers**, rated **4.7/5**.

Projects

loopview: open-source debugger for AI agents *github.com/YasmineZerai/loopview*

- Built and **published on PyPI** (`uvx loopview demo`) a tool that turns any **OpenTelemetry** trace into a **live graph** of an agent run: parallel agents, loops, handoffs, tool calls, with replay. One view for **LangGraph**, **Pydantic AI**, **CrewAI**, the **OpenAI Agents SDK** and Anthropic or OpenAI loops.
- Added a **cost tree** (where every token and dollar went) and a **Tools tab** that shows, across runs, how tools fail and what the agent did next. Python (FastAPI, SSE), **React** + TypeScript, tested on **real recorded traces**.

Multi-Agent Customer Support System (academic project, **team of 4**) *2025 - 2026*

- Built the **5 specialized agents** (order, complaint, FAQ, sales, escalation) under a **LangGraph** orchestrator, with **RAG** on **Pinecone**, each with **verify-before-act** rules: ownership checks, confirmation before cancelling, stock checks before recommending, silent fraud escalation.
- Fixed a complaint-agent failure that approved refunds on orders **46 days old** by splitting retrieval into **two ordered RAG searches** (eligibility, then resolution). **14 test scenarios across the 5 agents, all passing**.

Plant Disease Classification (deep learning, team of 4): **CNN** trained from scratch in **TensorFlow/Keras** on **54,305 images**, 38 classes, 36:1 imbalance. **95.72% accuracy**, **95.27% macro F1**. My error analysis exposed a worst class at **80.7% F1** hidden by the aggregate score.

Education

INSAT, National Institute of Applied Science and Technology *09/2022 - 05/2027*

Software Engineering, Engineering Degree (Master's equivalent), selective five-year program with admission by national competitive exam. Average **15/20**, ranked **20th** in my class.

Baccalaureate in Mathematics: 17.88/20, Highest Honors, national rank 297 of 8,097 (top 4%) *05/2022*

Technical Skills

AI / LLM: Python, **LangGraph**, **LangChain**, OpenAI API (function calling), **Anthropic SDK**, **Pydantic AI**, Gemini API, **RAG**, **ReAct** agents, multi-agent orchestration, tool design, embeddings, semantic chunking, **Pinecone**, **OpenTelemetry**, evaluation
Machine Learning & Data: Scikit-learn, **TensorFlow/Keras**, Pandas, NumPy, time-series data, EDA, feature engineering, model evaluation, **SQL**, **ETL pipelines**
Backend: FastAPI, async Python, Node.js, Nest.js, REST, WebSockets, Server-Sent Events, MQTT, **PostgreSQL**, MongoDB, Redis
Frontend: React, **TypeScript**, Next.js, Tailwind CSS, Recharts, React Flow
Infrastructure: Docker, **AWS** (EC2, S3), **GitHub Actions**, CI/CD, PyPI publishing, Linux, PostGIS. **Daily tools:** Claude Code, Cursor.
Languages: Arabic (native), French (C1), English (C1), Spanish (A2)